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Micro-Winkler titration method for dissolved oxygen concentration measurement

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ABSTRACT

In this report a gravimetric micro-Winkler titration method for determination of dissolved oxygen concentration in water is presented. Mathematical model of the method taking into account all influence factors is derived and an uncertainty analysis is carried out to determine the uncertainty contributions of all influence factors. The method is highly accurate: the relative expanded uncertainties ($k = 2$) are around 1% in the case of small (9–10 g) water samples. The uncertainty analysis carried out in characterizing the uncertainty of the method is the most comprehensive published for a micro-Winkler method, resulting in experimentally obtained estimates for all uncertainty sources of practical significance (around 20 uncertainty sources altogether).

1. Introduction

Dissolved oxygen (DO) is one of the most important substances present in water. Insufficient concentration of DO in natural waters leads to perishing of higher living organisms. High DO concentration is also very important for the efficiency of wastewater treatment plants. On the other hand oxygen is a strong oxidant and even a low DO concentration in, e.g. central heating systems may lead to premature repairs of the pipelines.

It is thus very important to be able to accurately measure DO concentration in water. The main methods of DO concentration measurement are the Winkler iodometric titration method (WM) [1,2], the amperometric sensor method [3] and now also the optical sensor method (based on luminescence quenching) [4,5].

Due to the speed and ease of use the amperometric method is the most widespread in routine analysis. At the same time it is a relative measurement method and needs calibration with the analyte and this means that these methods are not primary methods. DO is an unstable analyte and preparation and storage of calibration solutions is very difficult. Calibration is thus normally carried out in water saturated with air. Luminescence method is also a quick and simple method, very sensitive at low dissolved oxygen concentrations, but it is also a relative measurement method.

Although often perceived as a simple method, the physical and chemical processes on which the amperometric method is based are complex [6] and a large number of factors are involved in determining the result and its uncertainty. Among them are the saturation concentration itself, temperature, air composition, possible supersaturation, atmospheric pressure and humidity, stirring of the solution, stability of current, zero current value, drift, difference between calibration and measurement temperature (temperature compensation), condition of the sensor membrane, etc. [6]. The measurement uncertainty under expert laboratory conditions is in the range of ca ± 0.1–$0.2\ \mathrm{mg\,dm^{-3}}$ ($k = 2$) and under field conditions ca ± 0.2–$0.4\ \mathrm{mg\,dm^{-3}}$ [6]. The corresponding relative uncertainties (at saturation conditions at 25 °C) are in the ranges of 1.2–2.4% and 2.4–4.8%, respectively.

The published DO concentrations in water saturated with air [7] have been determined gasometry and WM [1]. Gasometry lacks chemical selectivity and therefore the WM can be considered the only truly primary method for DO determination. The WM method is based on the following. Two solutions are added to the oxygen-containing sample: one containing KI and KOH and the other containing $MnSO_4$. Oxygen reacts under alkaline conditions with Mn^{2+} ions forming manganese(III) hydroxide [8,9]:

$$4Mn^{2+} + O_2 + 8OH^- + 2H_2O \rightarrow 4Mn(OH)_3\downarrow \tag{1}$$

The solution is then acidified. Under acidic conditions Mn^{3+} ions oxidize iodide to iodine, which eventually forms I_3^- ions with the

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excess of KI [8,9]:

$$2Mn(OH)_3(s) + 3H_2SO_4 \rightarrow 2Mn^{3+} + 3SO_4{}^{2-} + 3H_2O \quad (2)$$

$$2Mn^{3+} + 2I^- \rightarrow 2Mn^{2+} + I_2 \quad (3)$$

$$I_2 + I^- \rightarrow I_3{}^- \quad (4)$$

The concentration of the formed $I_3{}^-$ ions is determined by titration with sodium thiosulfate solution, which is standardized using potassium iodate. Winkler method is the primary method of DO concentration measurement: the obtained DO concentration is traceable to the mass measurement of the standard substance KIO_3. Thus using the Winkler method it is possible to calibrate and test amperometric sensors with DO concentrations below saturation. This method also offers possibility of obtaining accurate DO concentration values for carrying out in situ interlaboratory comparison measurements [10].

Although DO determination by the classical WM, if executed by a skilled operator and with care, offers lower uncertainty and is less dependent on measurement conditions than the amperometric measurement, it is a demanding task that requires skill and is affected by numerous uncertainty sources. Several of these sources are specific to WM: contamination of the sample by atmospheric oxygen, oxygen dissolved in the reagents used, iodine volatilization, etc. We recently carried out an analysis of literature data and obtained the precision/accuracy of the classical WM performed by highly skilled operator as "standard deviation 0.1 mg dm^{-3}" [11]. This standard deviation combines both random and systematic effects and can be regarded as an estimate of the combined standard uncertainty of the WM in skilled hands. Results of interlaboratory comparisons as well as experience of practitioners suggest that the uncertainty under usual laboratory conditions is higher by a factor of 3–4 [12].

Different modifications of the original Winkler method have been proposed [8,13,14]. An important group of the modified methods are micro-methods [15–17] where sample volume is in the range of 1–10 cm^3 (samples of 100–150 cm^3 are used with the classical Winkler method). Several uncertainty sources become significantly more influential with micro-methods (sample volume around or below 10 cm^3). It is thus reasonable to expect that uncertainty of the micro-modifications of the WM tends to be higher that that of the classical WM, unless features are introduced that enhance accuracy. In our recent literature survey [11] we attempted to convert the accuracy and precision data from the original papers on micro WM-s [15–17] into uncertainty estimates compliant with the contemporary uncertainty paradigm [18]. It was not possible to do that entirely satisfactorily because of lack of information, but a $k = 2$ expanded uncertainty range of 0.2–0.3 mg dm^{-3} could reasonably describe these published procedures. The survey also indicated that all of these modifications were volumetric. Mass measurement, especially with small solution amounts and if syringes are used, is more accurate than volume measurement. This led us to the idea to use mass measurement instead of volume measurement for increasing the accuracy of the micro-Winkler method.

In this report we present a gravimetric modification of the Winkler micro-titration method, which uses inexpensive commercially available syringes and a conventional analytical balance. We demonstrate by validation the suitability of the method for determination of DO concentrations in the range of 2.9–12.4 mg dm^{-3} in water samples of 9–10 g. The expanded uncertainty of the method is in the range of ±0.08–0.14 mg dm^{-3} ($k = 2$). Thus this is a highly accurate (in terms of correctly estimated measurement uncertainty [11]) primary method for determination of DO concentration in small samples. We demonstrate that the uncertainty of the developed gravimetric method is more than two times lower than in the case of a similar volumetric method.

2. Experimental

In this section the developed gravimetric analysis method is described in detail, the calculation formulas are given, the sources of uncertainty and ways used for their estimation are listed. The procedure of analysis is described in Electronic Supplementary Material (ESM) with a series of photos. The volumetric method developed for comparison with the gravimetric method is described in the ESM.

2.1. Determination of the concentration of the $Na_2S_2O_3$ titrant

Iodine solution was prepared as follows. 2 cm^3 of the standard KIO_3 solution (0.0013 mol kg^{-1}) was transferred using a plastic syringe through plastic septum into a dried and weighed sample bottle. The bottle was weighed again. Using another syringe 0.1 cm^3 of solution containing KI (2.1 mol dm^{-3}) and KOH (8.7 mol dm^{-3}) (alkaline KI solution) was added. Using a third syringe ca 0.1 cm^3 of H_2SO_4 solution (2.7 mol dm^{-3}) was added carefully, until the color of the solution did not change anymore. Under acidic conditions iodine is formed according to the following reaction:

$$KIO_3 + 5KI + 3H_2SO_4 \rightarrow 3I_2 + 3K_2SO_4 + 3H_2O \quad (5)$$

The care in adding H_2SO_4 solution is necessary in order to avoid over-acidification of the solution because under strongly acidic conditions additional iodine may form via oxidation of iodide by air oxygen:

$$4I^- + 4H^+ + O_2 \rightarrow 2I_2 + 2H_2O \quad (6)$$

The iodine formed from KIO_3 was titrated immediately (to avoid loss of iodine by evaporation) with ca 0.0025 mol dm^{-3} $Na_2S_2O_3$ solution:

$$I_3{}^- + 2S_2O_3{}^{2-} \rightarrow 3I^- + S_4O_6{}^{2-} \quad (7)$$

It has been stressed [19] that loss of iodine may be an important source of uncertainty. This uncertainty has been estimated by us by additional experiments and has been taken into account as an uncertainty source (see the ESM for details).

Titration was carried out using a glass syringe filled with titrant and weighed. After titration the syringe was weighed again to determine the titrant mass. Six parallel measurements were carried out according to the described procedure and the average result was used as the titrant concentration.

2.2. Sample preparation

Samples were prepared in 10 cm^3 glass syringes with polytetrafluoroethylene (PTFE) plungers (Hamilton 1010LT 10.0 cm^3 Syringe, Luer Tip) (see the ESM for details). Masses of all syringes were determined beforehand.

Six parallel samples were taken as follows:

a) The syringe and the needle were rinsed with sample solution.
b) Air bubbles were eliminated by gently tapping the syringe. DO concentration decreases when doing this, therefore the syringe was emptied again so that only its dead volume was filled.
c) The syringe was rinsed again avoiding air bubbles. The concentration of oxygen in air per volume unit is more than 30 times higher than in water saturated with oxygen. Therefore avoiding air bubbles is extremely important.
d) 9.4 cm^3 of the sample was aspirated into the syringe.
e) The tip of the needle was poked into a rubber septum.

When six syringes were filled with samples and weighed the reagents were added. Ca 0.2 cm^3 of the alkaline KI solution and ca 0.2 cm^3 of $MnSO_4$ solution (2.1 mol dm^{-3}) was aspirated into each syringe. The needle tip was again sealed, the sample was intensely mixed (turning the syringe about ten times upside down until all the syringe was filled with the floating $Mn(OH)_3$ precipitate) and the $Mn(OH)_3$ precipitate was let to form during 45 ± 10 min (according to Eq. (1)). The syringe was weighed again to determine the net amount of the added reagents. This is necessary because the reagents also contain DO, which is taken into account. After 45 min ca 0.2 cm^3 of H_2SO_4 solution was aspirated into the syringe. Iodine is formed according to reactions (2), (3) and (4). At this stage the air bubbles do not interfere anymore.

2.3. Titration of the sample with the $Na_2S_2O_3$ titrant

The formed iodine solution is transferred through a plastic septum to the sample bottle. Simultaneously titrant is added from a pre-weighed glass syringe (to avoid possible evaporation of iodine). The sample syringe was rinsed twice with distilled water and the rinsing water was added to the sample bottle. The solution was titrated with $Na_2S_2O_3$ using a syringe until the solution was pale yellow. Then ca 0.2 cm^3 of 1% starch solution was added and titration was continued until the formed blue color disappeared. The titration syringe was weighed again. The amount of the consumed titrant was determined from mass difference. Six parallel titrations were carried out.

2.4. Determination of parasitic oxygen

Due to the small sample volume the possible sources of parasitic oxygen have to be considered. The two main sources of parasitic oxygen are the reagents and the PTFE plunger of the glass syringe.

The concentration of DO in the reagents is lower than in the sample and the amount of the reagents is small. Nevertheless the amount of oxygen introduced by the reagents can be around 1.5 μg and thus has to be taken into account. The overall amount of oxygen introduced by the $MnSO_4$ and the alkaline KI solutions was determined daily by aspirating into the glass syringe ca 2 cm^3 of the solution of KI and KOH, ca 2 cm^3 of $MnSO_4$ and the titration was carried out as described above.

All polymeric materials can dissolve oxygen. In this work the oxygen dissolved in the PTFE plunger is important. In order to determine the amount of oxygen introduced from the plunger the DO concentration of different quantities of the same sample were titrated. The mass of DO found in the sample was plotted against the sample mass. The mass of the parasitic oxygen introduced from the plunger was found as the intercept of the graph. The amount of parasitic oxygen introduced from the plunger was found as 2.11 μg with standard uncertainty of 0.27 μg. The $Mn(OH)_3$ precipitate was let to form during 45 ± 10 min in this experiment (see the ESM for details).

2.5. Mathematical model of the gravimetric method

Potassium iodate (KIO_3) was used as standard titrimetric substance. The stock solution concentration was found according to Eq. (8).

$$C_{KIO_3} = \frac{m_{KIO_3\_s} \cdot P_{KIO_3}}{M_{KIO_3} \cdot V_{flask} \cdot \rho_{KIO_3}} \tag{8}$$

where C_{KIO_3} [mol kg^{-1}] is the concentration of the KIO_3 solution, $m_{KIO_3\_s}$ [g] is the mass of the KIO_3, P_{KIO_3} [–] is the purity of KIO_3, M_{KIO_3} [g mol^{-1}] is molar mass of KIO_3, V_{flask} [dm^3] is the volume of the flask, ρ_{KIO_3} [kg dm^{-3}] is the density of 0.02847% KIO_3 solution (additional information is found in the ESM).

Concentration of the $Na_2S_2O_3$ titrant was found by titrating iodine liberated from the KIO_3 standard substance in acidic solution of KI. The titrant concentration was found according to Eq. (9).

$$C_{Na_2S_2O_3} = 6 \cdot C_{KIO_3} \cdot \Gamma^{m_{KIO_3}}_{m_{Na_2S_2O_3\_KIO_3}} \cdot F_{m_{KIO_3}} \cdot F_{m_{Na_2S_2O_3\_KIO_3}} \cdot F_{m_{KIO_3\_endp}} \cdot F_{I_2} \tag{9}$$

where $C_{Na_2S_2O_3}$ is the titrant concentration [mol kg^{-1}], m_{KIO_3} [g] is the mass of the KIO_3 solution taken for titration, $m_{Na_2S_2O_3\_KIO_3}$ [g] is the mass of the $Na_2S_2O_3$ titrant used for titrating the iodine liberated from KIO_3. $F_{m_{KIO_3}}$ [–] and $F_{m_{Na_2S_2O_3\_KIO_3}}$ [–] are factors taking into account the uncertainties of these solutions weighing. $F_{m_{KIO_3\_endp}}$ [–] is the factor taking into account the uncertainty of determining the titration end-point, F_{I_2} [–] is the factor taking into account evaporation of iodine from the solution. These factors have unity values and uncertainties corresponding to the relative uncertainties of the effects they account for. Value of $\Gamma^{m_X}_{m_Y}$ is calculated here and below by the general Eq. (10) and it is the average (of six parallel determinations) ratio of the amounts of X and Y solutions, used in the analysis.

$$\Gamma^{m_X}_{m_Y} = \frac{\sum_i m_{X\_i}/m_{Y\_i}}{6} \tag{10}$$

The concentration of parasitic DO in the reagents $C_{O_2\_reag}$ [mg kg^{-1}] was found as follows:

$$C_{O_2\_reag} = \frac{1}{4} \cdot M_{O_2} \cdot C_{Na_2S_2O_3} \cdot \Gamma^{m_{Na_2S_2O_3\_reag}}_{m_{reag}} \cdot F_{m_{reag}} \cdot F_{m_{Na_2S_2O_3\_reag}} \cdot F_{m_{reag\_endp}} \cdot F_{I_2} - \frac{O_{2\_syringe}}{m_{reag}} \tag{11}$$

where M_{O_2} [mg mol^{-1}] is the molar mass of oxygen, $m_{Na_2S_2O_3\_reag}$ [g] is the amount of titrant consumed for titration, m_{reag} [g] is the overall mass of the solutions of the alkaline KI and $MnSO_4$ and $O_{2\_syringe}$ [μg] is the mass of oxygen introduced by the syringe plunger.

F_{mreag} [–] and $F_{m_{Na_2S_2O_3\_reag}}$ [–] are factors taking into account the uncertainties of these solutions weighing. F_{mreag_endp} [–] is the factor taking into account the uncertainty of determining the titration end-point. These factors have unity values and uncertainties corresponding to the relative uncertainties of the effects they account for.

The DO concentration in the sample was found according to Eq. (12):

$$C_{O_2\_sample} = \rho \left(\frac{1}{4} \cdot M_{O_2} \cdot C_{Na_2S_2O_3} \cdot \Gamma^{m_{Na_2S_2O_3\_sample}}_{m_{sample}} \cdot F_{m_{sample}} \cdot F_{m_{Na_2S_2O_3\_sample}} \cdot F_{m_{sample\_endp}} \cdot F_{I_2} - C_{O_2\_reag} \cdot \Gamma^{m_{reag\_sample}}_{m_{sample}} - \frac{O_{2\_syringe}}{m_{sample}} \right) \tag{12}$$

where $C_{O_2\_sample}$ [mg dm^{-3}] is the DO concentration in the sample, ρ [kg dm^{-3}] is the density of water saturated with air (see the ESM for details), $m_{Na_2S_2O_3\_sample}$ [g] is the mass of $Na_2S_2O_3$ solution consumed for sample titration, m_{sample} [g] is the sample mass, m_{reag_sample} [g] is the overall mass of the added reagent solutions.

$F_{msample}$ [–] and $F_{m_{Na_2S_2O_3\_sample}}$ [–] are factors taking into account the uncertainties of these solutions weighing. $F_{msample\_endp}$ [–] is the factor taking into account the uncertainty of determining the titration end-point. These factors have unity values and uncertainties corresponding to the relative uncertainties of the effects they account for.

Uncertainty estimation has been carried out according to the ISO GUM modelling approach [20] using the Kragten spreadsheet method [21].

2.6. Quantifying the uncertainty components of the gravimetric method

The uncertainty estimation of MW gravimetric analysis was carried out on the basis of ISO GUM [20]. Measurements have been carried out in saturation conditions at 20 °C on the 22nd of February 2008. The calculations of measurement uncertainty are available in the ESM.

2.6.1. Uncertainty of concentration of KIO_3 solution

2.6.1.1. Specifying the measurand and defining the mathematical model. Potassium iodate (below KIO_3) solution with concentration of ca 0.0013 mol kg^{-1} was prepared in a 1000 cm^3 volumetric flask. 0.2847 g of KIO_3 (purity 99.7%) was weighed on a Mettler Toledo B204-S analytical balance (resolution 0.0001 g). The calibration uncertainty of the volumetric flask was ±0.4 cm^3 (information from producer). Concentration of the KIO_3 solution C_{KIO_3} [mol kg^{-1}] was found using Eq. (8). See the ESM for details.

2.6.1.2. Identification and evaluation of the uncertainty sources.

2.6.1.2.1. Mass of KIO_3, $m_{KIO_3\_s}$. Uncertainty of weighing solid KIO_3 has the following components: repeatability, u(repeatability) = 0.00016 g; rounding of the digital reading, u(rounding) = 0.000029 g; linearity of the balance, u(linearity) = 0.000115 g; and drift of the balance, u(drift) = 0.00024 g. The uncertainty contributions of these components have been found taking into account that KIO_3 mass was found as difference of masses of KIO_3 with tare and the empty tare. See ESM for more details.

2.6.1.2.2. Purity of KIO_3 standard substance, P_{KIO_3}. The certificate of KIO_3 has purity specified as min 99.7%. Assuming rectangular distribution we get 99.85% as the average purity and ±0.15% as the uncertainty, giving the relative standard uncertainty as 0.00087.

2.6.1.2.3. Molecular mass of KIO_3, M_{KIO_3}. Molecular mass and its uncertainty were found from atomic masses [22]: $M_{KIO_3} = 214.00097\ \mathrm{g\,mol^{-1}}$ and $u(M_{KIO_3}) = 0.00045\ \mathrm{g\,mol^{-1}}$.

2.6.1.2.4. Volume of the KIO_3 solution, V_{flask}. Uncertainty components of the 1 dm^3 volumetric flask volume are: uncertainty of the nominal volume as specified by the manufacturer (no calibration was done at our laboratory): ±0.4 cm^3 (u(cal) = 0.23 cm^3); uncertainty due to the imprecision of filling of the flask: ±10 drops or ±0.3 cm^3, u(filling) = 0.17 cm^3; uncertainty due to the temperature effect on solution density: u(temperature) = 0.24 cm^3. The standard uncertainty of the KIO_3 solution volume was found as $u(V_{\mathrm{flask}})$ = 0.38 cm^3.

2.6.1.3. Combined uncertainty of the KIO_3 solution concentration. $C_{KIO_3} = 0.0013304 \pm 0.0000036\ \mathrm{mol\,kg^{-1}}$, k = 2. The largest uncertainty contributions are due to weighing (47%) and purity of KIO_3 (42%).

2.6.2. Uncertainty of $Na_2S_2O_3$ concentration

2.6.2.1. Defining the measurand and the model. Concentration of the $Na_2S_2O_3$ solution was determined by titration of iodine formed from KIO_3 by titration with $Na_2S_2O_3$ solution. According to Eqs. (5) and (7) six moles of $Na_2S_2O_3$ is used for titration of iodine corresponding to one mole of KIO_3. Six repetitions were performed and ratios m_{KIO_3} and $m_{Na_2S_2O_3\_KIO_3}$ were found (see Eq. (9)). Concentration of $Na_2S_2O_3$ solution $C_{Na_2S_2O_3}$ [mol kg^{-1}] is the measurand.

2.6.2.2. Identification and evaluation of the uncertainty sources.

2.6.2.2.1. Concentration of the KIO_3 solution C_{KIO_3}. Uncertainty has been evaluated in Section 2.6.1.

2.6.2.2.2. Mass of KIO_3 solution m_{KIO_3} and mass of $Na_2S_2O_3$ titrant $m_{Na_2S_2O_3\_KIO_3}$. Uncertainty components for these masses are the same and are linearity and drift of the balance and rounding of the digital reading (see Section 2.6.1). These components are taken into account by F_{mKIO_3} and $F_{mNa_2S_2O_3\_KIO_3}$, respectively (see Eq. (9)). Repeatability is accounted for separately, by the uncertainty of the ratio of the masses $\Gamma^{m_{KIO_3}}_{m_{Na_2S_2O_3\_KIO_3}}$ (see below).

2.6.2.2.3. Uncertainty of the titration end-point. Random effects on the titration end-point are taken into account by the mean ratio m_{KIO_3} and $m_{Na_2S_2O_3\_KIO_3}$ (according to Eq. (10)). Possible systematic effects are taken into account by the factor $F_{mKIO_3\_endp}$ (see Eq. (9)). The end-point was determined using a visual indicator. The uncertainty of end-point determination was estimated as ±1 drop. Mass of one drop with the used syringe was 0.017 g and thus the standard uncertainty was u = 0.01 g. The average titrant consumption per titration was 9.5 g giving the relative standard uncertainty as 0.001.

2.6.2.2.4. Repeatability of m_{KIO_3} and $m_{Na_2S_2O_3\_KIO_3}$. Repeatabilities of masses are taken into account by the standard deviation of the mean ratio $\Gamma^{m_{KIO_3}}_{m_{Na_2S_2O_3\_KIO_3}}$ (according to Eq. (10)). Six repetitions were made. The mean ratio is 0.3095 and its standard deviation is 0.00028 giving the relative standard uncertainty of the ratio as 0.0009.

2.6.2.2.5. Evaporation of iodine before and during titration F_{I_2}. Titration of iodine was carried out in closed bottles and was performed quickly to minimize loss of iodine by evaporation. Some loss of iodine nevertheless can take place and the uncertainty due to this is taken into account by the factor F_{I_2} in the model (see Eq. (9)). Dedicated experiments were carried out in order to evaluate its uncertainty (see the ESM for details) giving the relative standard uncertainty as 0.00020.

2.6.2.3. Finding combined uncertainty of $C_{Na_2S_2O_3}$ concentration. Under our conditions the following was obtained: $C_{Na_2S_2O_3} = 0.002471 \pm 0.000010\ \mathrm{mol\,kg^{-1}}$, k = 2. The largest uncertainty contributors are the concentration of the KIO_3 solution (48%) and uncertainty due to the end-point determination (28%).

2.6.3. Uncertainty of DO concentration in the reagents (uncertainty of the blank)

2.6.3.1. Defining the measurand and the model. The measurand is the concentration of DO in the reagents $C_{O_2\_reag}$ [mg kg^{-1}] and the model is Eq. (11).

2.6.3.2. Identification and evaluation of the uncertainty sources.

2.6.3.2.1. Concentration of the $Na_2S_2O_3$ titrant $C_{Na_2S_2O_3}$. This concentration and its uncertainty have been evaluated in Section 2.6.2: 0.002471 ± 0.000010 mol kg^{-1}, k = 2.

2.6.3.2.2. Molar mass of oxygen M_{O_2}. The molar mass and its uncertainty were found based on data from literature [22]: $M_{O_2} = 31.9988\ \mathrm{g\,mol^{-1}}$ and $u(M_{O_2}) = 0.0003$.

2.6.3.2.3. Uncertainty of the net mass of reagent solutions m_{reag} and titrant mass $m_{Na_2S_2O_3\_reag}$. Uncertainty of these masses has the following components: Linearity and drift of the balance, rounding of the digital reading (see Section 2.6.1). These uncertainty components are taken into account by the factors F_{mreag} (see Eq. (11)). Repeatability of weighing is taken into account separately, by the standard deviation of the mean ratio of these masses $\Gamma^{m_{reag}}_{m_{Na_2S_2O_3\_reag}}$.

2.6.3.2.4. Uncertainty of the titration end-point. Possible systematic effect in finding titration end-point is taken into account by F_{mreag_end} (see Eq. (11)). This uncertainty has been estimated as ±2 drops. Mass of one drop is 0.017 g leading to the standard uncertainty of 0.02 g. Average titrant consumption for the blank titration was 0.9 g, giving the relative standard uncertainty as 0.022.

2.6.3.2.5. Repeatability of $m_{Na_2S_2O_3\_reag}$ and m_{reag}. Repeatability of the masses was taken into account by the standard deviation of the mean ratio $\Gamma^{m_{reag}}_{m_{Na_2S_2O_3\_reag}}$ (according to Eq. (10)). The mean ratio

$\Gamma^{m_{reag}}_{m_{Na_2S_2O_3\_reag}}$ is 0.1736 and its standard deviation is 0.0033 giving the relative standard uncertainty of the ratio as 0.019.

2.6.3.2.6. Evaporation of iodine before and during titration F_{I_2}. See Section 2.6.2.

2.6.3.2.7. Oxygen from the PTFE plunger, $O_{2\_syringe}$. Uncertainty due to oxygen diffusion into the solution from the PTFE plunger of the syringe was taken into account by the parameter $O_{2\_syringe}$ (see Eq. (11)). Dedicated experiments were carried out (see the ESM for details) and it was found that the mean (six repetitions) oxygen intake from the plunger is 2.11 μg and its standard deviation is 0.27 μg.

2.6.3.3. Combined uncertainty of DO concentration in the reagents. $C_{O_2\_reag} = 3.03 \pm 0.22$ mol kg^{-1}, $k = 2$. The largest uncertainty contribution is by determination of titration end-point (45%), followed by the repeatability of the six parallel measurements (34%).

2.6.4. Uncertainty of DO concentration in the sample

2.6.4.1. Defining the measurand and the model. The measurand is the concentration of DO in the sample $C_{O_2\_sample}$ [mg kg^{-1}] found according to Eq. (12).

2.6.4.2. Identification and evaluation of the uncertainty sources.

2.6.4.2.1. Concentration of the $Na_2S_2O_3$ titrant, $C_{Na_2S_2O_3}$. This uncertainty has been evaluated in Section 2.6.2: $C_{Na_2S_2O_3} = 0.00247 \pm 0.00001$ mol kg^{-1}, $k = 2$.

2.6.4.2.2. Mass of sample solution m_{sample} and mass of $Na_2S_2O_3$ titrant $m_{Na_2S_2O_3\_sample}$. Uncertainty of these masses has the following components: linearity and drift of the balance, rounding of the digital reading (see Section 2.6.1). These uncertainty components are taken into account by the factors $F_{m\mathrm{sample}}$ and $F_{m\mathrm{Na_2S_2O_3\_sample}}$ (see Eq. (12)). Repeatability of weighing is taken into account separately, by standard deviation of the mean ratio of these masses $\Gamma^{m_{sample}}_{m_{Na_2S_2O_3\_sample}}$ (see below).

2.6.4.2.3. Uncertainty of the titration end-point. Possible systematic effect in finding titration end-point is taken into account by $F_{m\mathrm{sample\_end}}$ (see Eq. (12)). This uncertainty has been estimated as ±1 drop. Mass of one drop is 0.017 g leading to the standard uncertainty of 0.01 g. Average titrant consumption for the sample titration was 4.3 g, giving the relative standard uncertainty as 0.0023.

2.6.4.2.4. Repeatability of $m_{Na_2S_2O_3\_sample}$ and m_{sample}. Repeatability of the masses was taken into account by the standard uncertainty of the mean ratio $\Gamma^{m_{sample}}_{m_{Na_2S_2O_3\_sample}}$ (according to Eq. (10)). The mean ratio is 0.4648 and the standard deviation of the mean is 0.00056 giving the relative standard uncertainty of the ratio as 0.0012.

2.6.4.2.5. Molar mass of oxygen, M_{O_2}. See Section 2.6.3.

2.6.4.2.6. Concentration of oxygen in the reagent solutions $C_{O_2\_reag}$. This has been evaluated in Section 2.6.3 as 3.03 ± 0.22 mol kg^{-1}, $k = 2$.

2.6.4.2.7. Repeatability uncertainty of m_{reag_sample} and m_{sample}. This uncertainty is taken into account as repeatability of the mean ratio of these masses $\Gamma^{m_{reag\_sample}}_{m_{sample}}$. The mean ratio of the masses of the reagents and the sample (according to Eq. (10)) is 0.0553 and the standard deviation of the mean is 0.0013 giving the relative standard uncertainty of the ratio as 0.023.

2.6.4.2.8. Evaporation of iodine before and during titration F_{I_2}. See Section 2.6.2.

2.6.4.2.9. Oxygen from the PTFE plunger, $O_{2\_syringe}$. See Section 2.6.3.

2.6.4.3. Combined uncertainty of DO concentration in the sample. The gravimetric MW analysis gave the following results: 8.78 ± 0.08 mg dm^{-3}, $k = 2$ (0.96%, relative). The largest uncertainty contribution was due to the correction of oxygen introduced from the plunger ($O_{2\_syringe}$) and end-point detection ($F_{m\mathrm{sample\_endp}}$) contributing 47% and 25%, respectively of the uncertainty, see Fig. 1.

Calculation of volumetric method is found in the ESM.

3. Results and discussion

3.1. Accuracy (measurement uncertainty) of the gravimetric method

The uncertainty budget of the developed gravimetric micro-Winkler method and of the volumetric method developed for reference are presented in Figs. 1 and 2, respectively. The uncertainty contributions differ slightly from measurement to measurement and the indicated contributions refer to one particular measurement.

The uncertainty of KIO_3 standard solution concentration accounts for 8.5% and 1.6% of the uncertainty in the gravimetric and volumetric method, respectively. In both methods the purity of KIO_3 and weighing of KIO_3 mass were the main contributors to uncertainty.

In contrast the largest uncertainty contributions to $C_{Na_2S_2O_3}$ are distinctly different in the case of gravimetric and volumetric methods. In the gravimetric method the main contributor is C_{KIO_3} (8.5% of the overall uncertainty). In the volumetric method the uncertainty of $C_{Na_2S_2O_3}$ is dominated by the volume-related uncertainties (accounted for by $F_{V\_KIO_3}$) and repeatability of titration $\Gamma^{V_{KIO_3}}_{V_{Na_2S_2O_3\_KIO_3}}$ (23% and 11%, respectively).

The contribution of the uncertainty due to the oxygen dissolved in the reagents is small: around 2% of the overall uncertainty in both methods.

The largest uncertainty contributions in the gravimetric method are uncertainty due to oxygen dissolved in the syringe plunger (47%) and uncertainty of end-point determination (25%). The main uncertainty contributions of the volumetric method are due to taking the reading (from syringe) during the two titrations involved, jointly contributing ($F_{V\_KIO_3}$, $F_{V\_Na_2S_2O_3\_sample}$, $F_{V\_Na_2S_2O_3\_KIO_3}$, F_{V_sample}) more than 50% of the uncertainty. All other uncertainty contributions are masked by these contributions and are at few per cent level. The dominance of the uncertainty sources related to volume measurement clearly indicates the superiority of gravimetric titration.

The uncertainty of the volumetric Winkler method with 9.4 cm^3 sample at saturation level is ±0.19 mg dm^{-3} (expanded uncertainty at $k = 2$ level). The uncertainty of the gravimetric micro-Winkler method in water at saturation level is more than two times lower: ±0.08 mg dm^{-3}.

3.2. Validation of the method

Both methods were validated by measuring DO concentration in water saturated with air. The concentrations of DO in water saturated with air were taken from the standard ISO 5814 [23]. Water was saturated with air at 5–25 °C using the procedure described earlier [7]. The results are given in Table 1.

The uncertainty takes into account the uncertainty of the tabulated value as well as the additional uncertainty due to the imperfections of the saturation conditions. The agreement between the reference value and the obtained results is very good: the absolute E_n values [24] are well below 1 in all cases. At the same time the results of the table indicate that the results of both methods are slightly (within uncertainty) biased. This bias may originate from the correction taking into account the oxygen diffusing into the syringe from the plunger: the negative bias indicates that this effect may be slightly “overcorrected”.

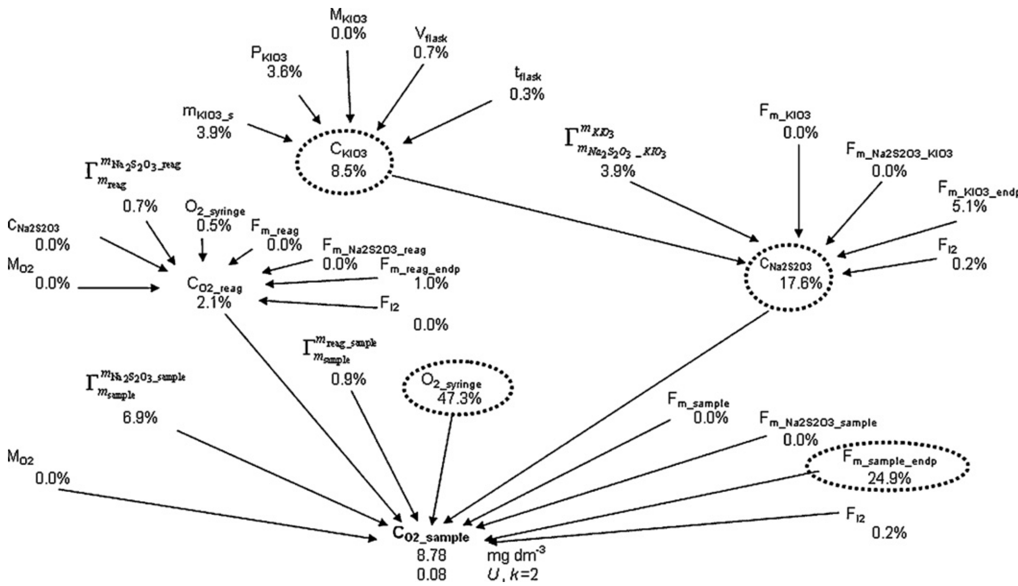

Fig. 1. Uncertainty budget of the gravimetric micro-Winkler method.

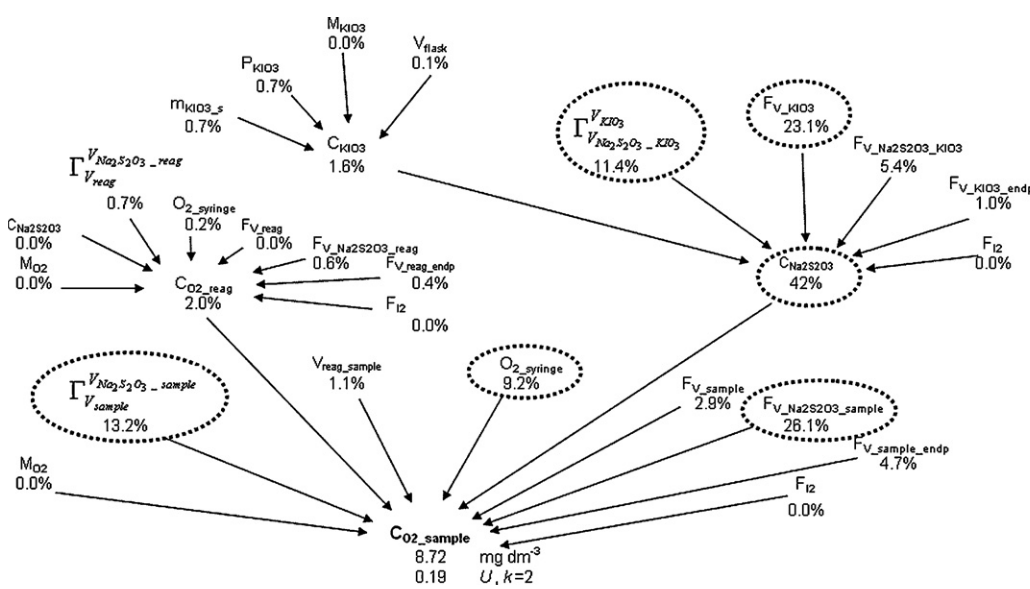

Fig. 2. Uncertainty budget of the volumetric micro-Winkler method (see the ESM for the definitions of the symbols).

Table 1
Results of the gravimetric and volumetric micro-Winkler methods (in mg dm^{-3}) under different experimental conditions.

Date	1.02.08	22.02.08	25.02.08	3.03.08	31.03.08	7.04.08	9.04.08	18.04.08
Saturation temperature	20 °C	20 °C	25 °C	5 °C	15 °C	20 °C	20 °C	20 °C
Reference value $C_{O_2\_saturation}$ [a]	9.20 ± 0.06	8.79 ± 0.06	8.10 ± 0.06	12.41 ± 0.08	10.17 ± 0.06	8.99 ± 0.06	8.92 ± 0.06	8.95 ± 0.06
Gravimetric micro-Winkler $C_{O_2\_sample}$ [a]	9.24 ± 0.09	8.78 ± 0.08	8.06 ± 0.10	12.31 ± 0.13	10.06 ± 0.13	8.97 ± 0.10	8.94 ± 0.14	8.96 ± 0.10
Δ [b]	0.04	−0.01	−0.04	−0.10	−0.11	−0.02	0.02	0.01
E_n [c]	0.4	−0.1	−0.3	−0.7	−0.7	−0.2	0.1	0.1
Volumetric micro-Winkler $C_{O_2\_sample}$ [a]		8.72 ± 0.19	8.02 ± 0.18	12.30 ± 0.24	10.14 ± 0.26			
Δ [b]		−0.07	−0.08	−0.11	−0.03			
E_n [c]		−0.4	−0.4	−0.4	−0.1			

[a] All values are given with $k = 2$ uncertainties.
[b] $\Delta = C_{O_2\_sample} - C_{O_2\_saturation}$.
[c] The E_n values are calculated and interpreted as explained in ref [24]: $|E_n| \leq 1$ means agreement, $|E_n| > 1$ means disagreement.

The gravimetric method was validated also by measuring DO concentration in water saturated with mixture of air and nitrogen. The obtained concentrations with estimated uncertainty were $2.90 \pm 0.10\,mg\,dm^{-3}$ ($k=2$) and $5.10 \pm 0.12\,mg\,dm^{-3}$ ($k=2$). These results were compared with amperometric method (WTW Oxi340i with CellOx 325 sensor). Uncertainties were calculated [6] and concentration of dissolved oxygen was found as $2.85 \pm 0.12\,mg\,dm^{-3}$ ($k=2$) and $5.07 \pm 0.18\,mg\,dm^{-3}$ ($k=2$), respectively. The calculated E_n values [24] are 0.3 and 0.1, respectively, indicating good agreement.

4. Conclusions

As a result of this work an accurate gravimetric micro-Winkler method for determination of DO concentration has been developed. The method is highly accurate: its expanded uncertainty is in the range of 0.08–0.14 $mg\,dm^{-3}$.

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Appendix A. Supplementary material

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.aca.2009.06.067.

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